

Practical Project: Evaluating Multi-View Video Diffusion for Humanoid Mesh Animation

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September 10, 2026

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Abstract

Recent approaches to mesh animation use video diffusion models to generate video-based motion priors from which explicit 3D motion is subsequently reconstructed. However, recovering motion from a single view is inherently underconstrained due to depth ambiguities and self-occlusions. This project investigates whether a pretrained multi-view video diffusion model can synthesize additional viewpoints that provide useful geometric constraints for downstream 3D human motion reconstruction. In contrast to existing multi-view mesh animation approaches, we focus specifically on human meshes, enabling the use of human-specific reconstruction methods and low-dimensional body priors. We propose a general framework for human mesh animation and evaluate an experimental instantiation based on SV4D 2.0 and an EasyMocap-based reconstruction pipeline in a controlled setting on CAPE motion sequences. Our results show that synthesized target views substantially improve reconstruction accuracy over single-view baselines, reducing MPVPE to 58.55 mm compared with 83.98 mm for Multi-HMR and 112.57 mm for WHAM. Reconstruction performance depends on both the angular placement and number of synthesized viewpoints. Conditioning SV4D 2.0 on target-view renderings of the input mesh further improves synthesized-view fidelity and downstream reconstruction accuracy. Finally, we address long-horizon generation by introducing a mesh-level reconditioning strategy that uses intermediate 3D reconstructions to reduce error accumulation. While this improves synthesized-view fidelity over the native continuation of SV4D 2.0, reconstruction accuracy still degrades over longer horizons. Overall, the results demonstrate the potential of multi-view video diffusion for human mesh animation while identifying reliable long-horizon generation as a key remaining challenge.

1 Introduction

1.1 Problem Statement

Animating humanoid 3D models is a fundamental problem in computer graphics, with applications ranging from content creation to video games and virtual reality. Recent advances in deep learning have enabled increasingly powerful data-driven approaches for human motion generation [1]. A central challenge lies in generating realistic and semantically meaningful motion for a given human mesh while preserving the identity and geometry of the target character.

Established learning-based motion generation methods are typically trained directly on three-dimensional motion representations such as sequences of joint positions or joint rotations [2]. While such approaches can generate plausible motion, they depend on suitable 3D motion data for training, which

is costly to acquire and difficult to scale [1]. Moreover, the majority of these approaches generate motion independently of the geometry of a specific target character. As a result, animating a given input mesh typically requires an additional motion-transfer or retargeting step [3].

An alternative is to represent motion indirectly through a video-based prior. Modern video diffusion models (VDMs) can generate temporally coherent visual motion from semantic and image-based conditioning [4, 5] and can therefore serve as motion priors without being trained directly on 3D motion-capture data. When conditioned on a rendering of a given human model, the generated video can depict the desired motion directly with the appearance of the target character. An explicit 3D animation can subsequently be recovered from this video [6].

However, recovering 3D motion from a single-view video introduces inherent geometric ambiguities. Depth is not directly observable from monocular imagery, and body parts may become self-occluded throughout the motion. Consequently, different 3D configurations can result in similar monocular observations, making the subsequent reconstruction problem underconstrained [7, 6].

1.2 Motivation

Multi-view video diffusion provides a potential way to reduce these ambiguities. Starting from a monocular reference video, a multi-view video diffusion model (MV-VDM) can synthesize synchronized observations of the same dynamic content from additional viewpoints [8]. If these synthesized views are sufficiently consistent, they provide complementary observations of the subject and can introduce additional geometric constraints during 3D reconstruction.

While initial work has already explored multi-view video diffusion for mesh animation, these approaches target general 3D models rather than humans specifically [9, 10]. Focusing on human meshes enables the use of human-specific reconstruction methods and low-dimensional body priors, providing additional constraints for recovering the underlying 3D motion [7]. Furthermore, existing approaches rely on generative models specifically designed for their animation setting [9, 10]. In contrast, this work investigates whether a pretrained general-purpose MV-VDM can be combined with an independent human-specific reconstruction pipeline for human mesh animation.

This motivates the central question investigated in this project: whether synthesized target-view videos can provide sufficiently useful and consistent observations to improve 3D human motion reconstruction within a motion generation pipeline compared with relying on a single view alone. Beyond this central question, the effectiveness of the proposed pipeline may depend strongly on its configuration. We therefore investigate how factors such as the

angular spacing and number of target viewpoints, as well as the availability of target-view conditioning information, affect synthesized-view fidelity and downstream 3D reconstruction accuracy.

A further challenge is the limited temporal generation window of current multi-view video diffusion models. Generating longer animations therefore requires extending the synthesis beyond this native window, for example through autoregressive continuation, which can propagate and accumulate synthesis errors over time. We consequently investigate alternative reconditioning strategies and evaluate whether incorporating an explicit 3D reconstruction step can improve long-horizon generation and downstream motion reconstruction.

1.3 Project Goals

The primary objectives of this project are:

- To develop a general architecture for human mesh animation that uses a video-based motion prior, together with multi-view synthesis and human-specific 3D reconstruction.
- To investigate whether synthesized target-view videos provide useful additional constraints for 3D human motion reconstruction compared with single-view reconstruction.
- To analyze how different pipeline configurations and components influence synthesized-view quality and downstream reconstruction accuracy.
- To investigate how multi-view video generation can be continued beyond the model’s native generation window and to evaluate suitable strategies for longer animation sequences.

1.4 Project Scope

This project focuses on evaluating the use of multi-view video diffusion for human mesh animation. While the proposed architecture describes a complete pipeline from a static human input mesh and an action description to an animated mesh sequence, the experimental evaluation concentrates on the multi-view synthesis and reconstruction stages. To isolate these components, known motion sequences are used to provide controlled reference videos instead of generating them with an additional single-view VDM. This also enables quantitative evaluation against known 3D ground-truth motion.

The proposed pipeline is developed as a research prototype for investigating the feasibility and limitations of this approach rather than as a

deployment-ready animation system. Its implementation is therefore designed primarily to enable controlled experimentation and evaluation of the individual pipeline components.

In the experimental setup, the reconstruction stage recovers motion in the SMPL representation. However, the overall architecture is not inherently restricted to SMPL and could be extended to arbitrary human meshes by using registration and motion transfer to map the reconstructed motion back to the original input geometry. Such an extension is considered conceptually in this work, but is not implemented or evaluated as part of the project.

1.5 Report Structure

The remainder of this report is organized as follows: Section 2 introduces the required background and reviews related approaches to human motion generation. Section 3 first presents the proposed system architecture in a generic form and then describes its concrete experimental implementation. Section 4 presents the experimental evaluation and addresses the research questions defined for this project. Finally, Section 5 summarizes the main findings, discusses limitations, and outlines directions for future work.

2 Background and Related Work

This chapter is divided into two sections. The first introduces the technical preliminaries required for the proposed approach (Section 2.1). The second provides an overview of related work in human motion generation and positions our work within this research landscape (Section 2.2).

2.1 Preliminaries

2.1.1 SMPL Human Body Model

SMPL [11] is a parametric human body model that represents human body geometry through a low-dimensional set of pose and shape parameters. The shape parameters describe variations in proportions and physique, while the pose parameters describe the articulation of the underlying skeleton. Given these parameters, SMPL produces a 3D human mesh with fixed topology, such that different body shapes and poses are represented using the same underlying mesh structure.

This compact and consistent representation makes SMPL well suited for human motion modeling and reconstruction. A motion sequence can be represented by varying the body pose, global orientation and translation over time while maintaining a consistent body shape throughout the sequence.

2.1.2 Video Diffusion Models

Video diffusion models [4] extend diffusion-based generative modeling [12] from individual images to video sequences by jointly modeling spatial appearance and temporal evolution. Starting from randomly sampled noise, an iterative denoising process produces a sequence of video frames by modeling dependencies across both spatial and temporal dimensions.

Video generation can additionally be conditioned on information such as text or images [5]. In image-conditioned video generation, an input image provides the appearance and initial configuration of the subject, while the model synthesizes its temporal evolution over subsequent frames. Although VDMs do not represent motion explicitly through 3D poses or mesh deformations, the temporal evolution of the generated visual content implicitly encodes motion.

2.1.3 Multi-View Video Diffusion Models

Multi-view video diffusion models extend conventional video diffusion by jointly generating videos of the same dynamic content from multiple viewpoints [13, 8, 14, 15]. In addition to maintaining temporal consistency within each video, the generated videos must remain consistent across viewpoints such that appearance and motion correspond across different camera perspectives. Maintaining this consistency simultaneously across the temporal and viewpoint dimensions remains a central challenge of multi-view video diffusion [13, 8].

Depending on the model and application, multi-view video generation can be conditioned on different forms of visual and camera information. These may include a single reference image or a monocular reference video, together with camera poses or trajectories specifying the desired viewpoints [8, 14, 15]. The conditioning therefore determines which aspects of the dynamic scene are provided explicitly and which must be inferred by the model.

2.2 Human Motion Generation

Existing approaches to human motion generation can broadly be distinguished by the representation on which the generative model is trained. Most existing methods operate on what we refer to in this work as a *direct motion representation*: they are trained directly on 3D motion data, such as sequences of joint positions, joint rotations, or parameters of a parametric human body model and generate motion in the corresponding representation.

In contrast, methods based on what we refer to as an *indirect motion representation* use a video-based motion prior rather than learning motion directly from 3D motion data. These approaches employ a VDM trained on video data to synthesize a sequence in which motion is represented implicitly

through the temporal evolution of the depicted subject. An explicit 3D motion representation is subsequently recovered from the generated video.

Besides the motion representation itself, an important consideration for the goal of animating a given input mesh is how the geometry of the target character is incorporated. Ideally, the approach should be applicable to arbitrary human input meshes rather than being restricted to a fixed body shape or representation. This requires either adapting a generated motion to the geometry of the input mesh or incorporating information about the target geometry directly into the generation process.

In the following, approaches based on direct motion representations (Section 2.2.1) and indirect motion representations (Section 2.2.2) are reviewed, with particular emphasis on their suitability for animating arbitrary human input meshes.

2.2.1 Direct Motion Representation

Approaches based on a direct motion representation train generative models directly on three-dimensional motion data, typically obtained through motion capture (MoCap). Human motion is represented as sequences of skeletal poses, for example through joint positions, joint rotations, or parameters of a parametric human body model. For semantically conditioned motion generation, datasets such as KIT Motion-Language [16], BABEL [17], and HumanML3D [18] provide textual or action-based annotations for such 3D motion sequences. Existing approaches can further be distinguished by whether motion is generated independently of a specific target body or whether the geometry of the target character is incorporated during generation.

Shape-Agnostic Generation. A large body of work generates generic human motion without conditioning on a specific body shape or input mesh.

One established line of work employs autoencoder-based architectures to learn compact latent representations of 3D motion. Action2Motion [19] introduced a variational autoencoder-based formulation for motion generation, synthesizing motion conditioned on an action class. TEMOS [20] extends this idea to free-form text by learning a shared latent space between textual descriptions and motion sequences. MotionCLIP [21] similarly learns a latent motion representation, but aligns it with the pretrained CLIP embedding space to exploit its semantic knowledge. T2M-GPT [22] instead uses a VQ-VAE to encode motion into discrete representations and subsequently generates the corresponding code sequence autoregressively from a text prompt. However, the generalization of purely autoencoder-based approaches remains limited, particularly when motions differ substantially from those represented in the training data [1].

Diffusion models have subsequently emerged as a promising alternative to autoencoder-based motion generation. MotionDiffuse [23] was the first

approach to apply diffusion-based generation to text-conditioned human motion, generating motion through an iterative denoising process conditioned on textual embeddings. Subsequent work further developed this paradigm. MDM [24], for example, employs a transformer-based architecture and directly predicts the motion sample at each denoising step, enabling the use of additional geometric constraints during training.

However, the generated motion is generally independent of the geometry of a specific target character. It can subsequently be transferred to another skeleton or body model through motion retargeting [3]. Nevertheless, such post-hoc transfer generally does not account for the fact that the execution of a motion can depend on the proportions and geometry of the target body.

Shape-Conditioned Generation. Shape-conditioned approaches address this limitation by incorporating information about the target body directly into motion generation. HUMOS [25] conditions motion on body-shape and identity-related features, although its primary focus is the adaptation of existing motion rather than semantic text- or action-conditioned generation. SMD [26] generates mesh sequences conditioned jointly on a target mesh and a semantic motion description, using a compact spectral representation to encode the input geometry. TapMo [27] further extends shape-aware generation to arbitrary static meshes by predicting deformation handles and skinning weights and conditioning motion generation on deformation features derived from the input mesh. In this way, the geometry of the target character can influence the generated motion rather than being considered only during subsequent retargeting.

Despite these advances, shape-conditioned approaches based on direct motion representations remain comparatively underexplored. A central limitation is their dependence on suitable 3D motion data for training. MoCap data is costly to acquire and difficult to scale, which restricts the diversity of available training data [1]. This limitation is particularly relevant for shape-conditioned generation, which requires training data that captures both diverse motions and variations in body shape in order to learn their relationship [25].

2.2.2 Indirect Motion Representation

Approaches based on an indirect motion representation avoid training the generative model directly on 3D motion data and instead exploit video-based motion priors generated by pretrained VDMs. Compared with MoCap data, video data can be collected at substantially larger scale, providing generative models with a broader and more diverse source of motion information [5]. The resulting video does not contain an explicit 3D motion representation, but encodes motion implicitly through the temporal evolution of the depicted

subject. A subsequent motion-reconstruction stage then recovers an explicit 3D motion representation from the generated video [7, 28].

For the animation of arbitrary input meshes, this formulation is particularly attractive because the geometry and appearance of a given input mesh can be incorporated through visual conditioning. A rendering of the input mesh can be provided to the VDM, allowing the generated motion prior to depict the target character directly. The recovered motion can then be transferred back to the corresponding 3D geometry. Several existing approaches therefore operate on arbitrary input meshes rather than being restricted to a fixed body shape or representation [9, 28, 7].

Single-View Approaches. Single-view approaches employ a conventional VDM to generate a motion prior from a single reference viewpoint, typically conditioned on a rendering of the input mesh and a textual motion description. The generated video is then used to recover the corresponding 3D animation through visual tracking or optimization. MotionDreamer [6], one of the more prominent approaches in this category, tracks semantic features extracted from the employed VDM and optimizes the motion of the input mesh accordingly. Puppeteer [28] combines automatic rigging with several rendering- and tracking-based objectives to optimize the joint motion of arbitrary input meshes. Animating the Uncaptured [7] specifically targets humanoid meshes. The static input mesh is first registered to an SMPL model that serves as a deformation proxy. During motion recovery, a learned human body prior is used to constrain the estimated poses, while silhouette, body-landmark, and dense semantic features provide the visual tracking signal.

A key limitation of these approaches is that the 3D motion must be recovered from a single viewpoint. Occlusions and depth ambiguities can therefore make the underlying 3D configuration difficult to infer and may introduce errors when transferring the observed motion back to the 3D representation. Although body priors and additional regularization can reduce these ambiguities, the missing information from unobserved viewpoints remains inherent to the single-view setting [7, 6].

Multi-View Approaches. Multi-view approaches address the ambiguities of single-view motion reconstruction by synthesizing the same motion from multiple target viewpoints and using the resulting videos jointly for 3D reconstruction. The additional perspectives provide complementary information about body parts that may be occluded in individual views and reduce ambiguities in recovering the underlying 3D motion. At the same time, multi-view generation introduces the additional requirement that the synthesized videos remain consistent both over time and across viewpoints, which remains a central challenge for MV-VDMs [8].

Research on multi-view approaches is still relatively recent, and only a limited number of studies have explored their use. Animate3D [10] employs a dedicated MV-VDM conditioned on renderings of an arbitrary static 3D model. The synthesized target-view videos are first used to reconstruct the underlying motion and are subsequently refined through 4D Score Distillation Sampling (SDS). AnimaX [9] employs a purpose-built joint video-pose diffusion model that generates videos from multiple viewpoints together with corresponding pose information. Images and pose maps rendered from the input mesh are used as conditioning, and joint positions are subsequently triangulated across the synthesized viewpoints before being converted into the final animation.

The approach investigated in this work likewise follows a multi-view video-based formulation and is therefore related to the approaches discussed above. Unlike these approaches, however, it specifically considers the animation of human meshes. Restricting the problem to humans enables the use of human-specific reconstruction methods and low-dimensional body priors, similar to the SMPL-based formulation employed in [7]. At the same time, the proposed pipeline does not rely on a purpose-built generative model for mesh animation. Instead, it investigates whether a pretrained general-purpose MV-VDM can be combined with an independent multi-view human mesh reconstruction pipeline. Motion recovery relies on image-derived information from the synthesized target-view videos, without requiring additional pose information generated by the diffusion model as in AnimaX [9] or subsequent SDS-based refinement as in Animate3D [10]. The resulting setup is designed to investigate whether the visual consistency provided by a general-purpose MV-VDM is sufficient to support accurate downstream human motion reconstruction and to what extent the additional synthesized viewpoints improve reconstruction compared with a single-view setting.

3 Method

This chapter presents the proposed approach to human mesh animation using video-based motion priors and multi-view video synthesis. Section 3.1 first introduces the overall system architecture at an abstract level and formalizes its individual stages. Section 3.2 then describes the concrete instantiation and experimental setup used in this work.

3.1 System Architecture

The proposed architecture aims to generate a mesh sequence $\{\widehat{M}_t\}_{t=0}^{T-1}$ from a static human input mesh $M^{\text{in}} \in \mathcal{M}$ and an action description $a \in \mathcal{A}$, where $\mathcal{M}$ denotes the space of human meshes, $\mathcal{A}$ the space of action descriptions, and T the number of frames in the generated sequence. The resulting mesh sequence should represent the motion specified by a while preserving the

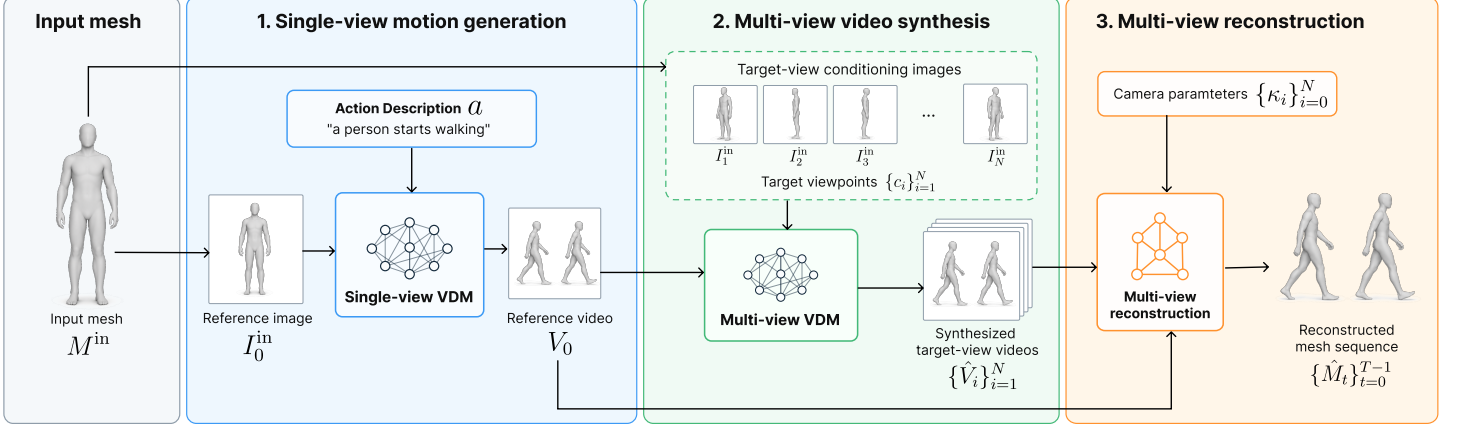


Figure 1: Overview of the proposed system architecture for human mesh animation.

identity of the input mesh M^{in} . An overview of the proposed architecture is shown in Figure 1.

The architecture builds on existing single-view animation pipelines, in particular the method of Benedí San Millán et al. [7]. Up to the generation of a single reference video, the proposed approach follows the same general principle: a rendering of the input mesh together with the action description is used to generate a video-based motion prior (Section 3.1.1). Instead of directly reconstructing the animation from this monocular video, however, we introduce an additional multi-view video synthesis stage (Section 3.1.2). The resulting synchronized target-view videos are then jointly used with the reference video for multi-view reconstruction (Section 3.1.3).

For simplicity, we assume throughout the following that the input mesh M^{in} is represented by an SMPL model. Applying the architecture to arbitrary human meshes requires additional steps, which are discussed in Section 3.1.5.

3.1.1 Single-View Motion Generation

The first stage follows the general principle of existing approaches based on indirect motion representation. Instead of generating motion directly in 3D space, the desired motion is first represented by a video-based motion prior.

The input mesh M^{in} is therefore rendered from a reference viewpoint $c_0 \in \mathcal{C}$, where $\mathcal{C}$ denotes the set of considered viewpoints. Let $\mathcal{I}$ denote the space of RGB images and $\mathcal{P} : \mathcal{M} \times \mathcal{C} \rightarrow \mathcal{I}$ the corresponding rendering operator. The reference image is then given by

$$I_0^{\text{in}} = \mathcal{P}(M^{\text{in}}, c_0). \quad (1)$$

The reference image I_0^{in} and the action description a are subsequently used to condition a single-view VDM. Let $\mathcal{V}_T = \mathcal{I}^T$ denote the space of

videos containing T frames, and let $\mathcal{S} : \mathcal{I} \times \mathcal{A} \rightarrow \mathcal{V}_T$ denote the single-view VDM. The resulting reference video is given by

$$V_0 = \mathcal{S}(I_0^{\text{in}}, a). \quad (2)$$

The resulting reference video provides an indirect representation of the motion specified by the action description while aiming to preserve the visual characteristics of the input mesh.

3.1.2 Multi-View Video Synthesis

The objective of the multi-view video synthesis stage is to generate temporally synchronized videos of the motion contained in the reference video V_0 from a set of target viewpoints $\{c_i\}_{i=1}^N$.

Since the input mesh M^{in} is known, it can additionally be rendered from each target viewpoint to provide view-specific appearance information for conditioning the MV-VDM. These renderings therefore introduce complementary information that is not available in the monocular reference video V_0 . The corresponding target-view images are defined as

$$I_i^{\text{in}} = \mathcal{P}(M^{\text{in}}, c_i), \quad i = 1, \dots, N. \quad (3)$$

We represent the multi-view video diffusion model by the operator $\mathcal{G} : \mathcal{V}_T \times \mathcal{I}^N \times \mathcal{C}^N \rightarrow \mathcal{V}_T^N$. Given the reference video, the target-view renderings, and the corresponding target viewpoints, the MV-VDM generates a set of synthesized target-view videos according to

$$\{\widehat{V}_i\}_{i=1}^N = \mathcal{G}(V_0, \{I_i^{\text{in}}\}_{i=1}^N, \{c_i\}_{i=1}^N). \quad (4)$$

Each synthesized target-view video $\widehat{V}_i = \{\widehat{I}_{t,i}\}_{t=0}^{T-1}$ approximates how the motion contained in the reference video would be observed from viewpoint c_i . Together, the reference video and the synthesized target views form a synchronized multi-view representation of the motion.

3.1.3 Multi-View Reconstruction

The final stage converts the indirect multi-view motion representation back into an explicit 3D mesh sequence. For this purpose, the reference video V_0 and the synthesized target-view videos $\{\widehat{V}_i\}_{i=1}^N$ are jointly used for multi-view reconstruction.

In addition to the videos, the corresponding camera parameters are required. We denote the camera parameters associated with viewpoint c_i by $\kappa_i \in \mathcal{K}$, where $\mathcal{K}$ denotes the space of camera parameters required by the reconstruction method. Let $\mathcal{R} : (\mathcal{V}_T \times \mathcal{K})^{N+1} \rightarrow \mathcal{M}^T$ denote the multi-view reconstruction operator. The reconstructed mesh sequence is then given by

$$\{\widehat{M}_t\}_{t=0}^{T-1} = \mathcal{R}\left((V_0, \kappa_0), \{(\widehat{V}_i, \kappa_i)\}_{i=1}^N\right). \quad (5)$$

By using multiple synchronized videos, the reconstruction incorporates additional geometric constraints, which can potentially reduce the geometric ambiguities inherent in reconstructing 3D motion from a single monocular video. The resulting sequence $\{\widehat{M}_t\}_{t=0}^{T-1}$ constitutes the final output of the proposed architecture.

3.1.4 Long-Horizon Generation

If the employed multi-view VDM $\mathcal{G}$ operates on a limited temporal generation horizon, longer motions require repeated application of the synthesis process to consecutive generation windows. We therefore divide the reference video into S consecutive, overlapping generation windows with temporal boundaries

$$0 = \tau_0 < \tau_1 < \dots < \tau_S = T - 1, \quad (6)$$

where generation window s contains the frames $t = \tau_s, \dots, \tau_{s+1}$. Consequently, adjacent windows share the boundary frame at τ_s for $s > 0$, which is used to condition the subsequent window. Let $V_0^{(s)}$ denote the corresponding window of the reference video and $\widehat{V}_i^{(s)} = \{\widehat{I}_{t,i}^{(s)}\}_{t=\tau_s}^{\tau_{s+1}}$ the synthesized generation window for target viewpoint c_i .

For the first generation window, the original target-view renderings $\{I_i^{\text{in}}\}_{i=1}^N$ are used for conditioning. For subsequent windows, however, these renderings still represent the initial pose of the input mesh and therefore no longer correspond to the current motion state. Suitable target-view conditioning must consequently be obtained from the preceding generated window.

The most direct strategy is *frame-level autoregressive reconditioning*. For each target viewpoint, the final synthesized frame of the preceding generation window is reused as the conditioning frame at the shared temporal boundary. For $s > 0$, the synthesis is therefore given by

$$\{\widehat{V}_i^{(s)}\}_{i=1}^N = \mathcal{G}\left(V_0^{(s)}, \{\widehat{I}_{\tau_s,i}^{(s-1)}\}_{i=1}^N, \{c_i\}_{i=1}^N\right). \quad (7)$$

This reconditioning step can be applied repeatedly across consecutive generation windows without requiring any additional processing. However, synthesis errors or visual distortions contained in the generated boundary frames are directly propagated to subsequent windows and may accumulate over longer generation horizons.

As an alternative, we propose *mesh-level autoregressive reconditioning*, which can replace frame-level reconditioning at selected window boundaries. After completing the preceding generation window, its synthesized target-view videos $\{\widehat{V}_i^{(s-1)}\}_{i=1}^N$ are reconstructed together with the corresponding reference-video window $V_0^{(s-1)}$ using the reconstruction operator $\mathcal{R}$. At the

shared temporal boundary τ_s , the resulting reconstructed mesh $\widehat{M}_{\tau_s}^{(s-1)}$ is rendered from each target viewpoint and used instead of the synthesized boundary frames to condition the following window. For such a boundary, the synthesis is given by

$$\{\widehat{V}_i^{(s)}\}_{i=1}^N = \mathcal{G}\left(V_0^{(s)}, \{\mathcal{P}(\widehat{M}_{\tau_s}^{(s-1)}, c_i)\}_{i=1}^N, \{c_i\}_{i=1}^N\right), \quad s > 0. \quad (8)$$

By projecting the generated motion into an explicit 3D representation before reconditioning, this strategy is intended to reduce the direct propagation of visual errors across consecutive generation windows.

3.1.5 Applicability to Arbitrary Human Meshes

The described architecture assumes that the input mesh M^{in} is represented by a parametric human body model. In this work, we use SMPL for this representation. While this allows the pipeline to operate on different human body shapes, arbitrary surface geometry such as clothing or accessories cannot be represented directly.

To apply the architecture to an arbitrary human mesh, an SMPL model can be used as a deformation proxy, following the approach proposed in [7]. The SMPL model is first registered to the input mesh by optimizing its scale, shape, initial pose, translation, and rotation parameters to match the input geometry. Each vertex of the input mesh is then anchored to its closest face on the fitted SMPL model, establishing a correspondence between both representations. Motion can subsequently be reconstructed in SMPL parameter space and transferred back to the original mesh through this correspondence. These additional steps are outside the scope of this work.

3.2 Experimental Implementation

While Section 3.1 introduced the proposed architecture at an abstract level, this section describes its concrete instantiation for the conducted experiments. Rather than evaluating the complete pipeline end-to-end, we focus on the multi-view synthesis and multi-view reconstruction stages and investigate their contribution to the resulting 3D motion reconstruction.

The experimental pipeline was implemented in Python. Multi-view video synthesis with SV4D 2.0 was performed on NVIDIA GeForce RTX 4090 and RTX 3090 GPUs with 24 GB of VRAM, while the EasyMocap-based reconstruction pipeline was additionally executed on an NVIDIA GeForce RTX 2080 Ti with 12 GB of VRAM. Mesh rendering was performed using Trimesh 4.12.2 and Pyrender 0.1.45. Further implementation details, including the software environments used for the individual processing stages, are provided in Appendix A.1

3.2.1 Controlled Experimental Setup

To isolate the multi-view synthesis and reconstruction stages from errors introduced by upstream single-view motion generation, the single-view VDM $\mathcal{S}$ is not instantiated in the experimental setup. Instead, known MoCap mesh sequences are used as controlled inputs to the remaining pipeline stages. This enables a quantitative evaluation against known 3D ground-truth motion, which would not be available for fully generated end-to-end sequences. We denote such a source mesh sequence by $\{M_t^{\text{src}}\}_{t=0}^{T-1}$.

Using the rendering operator $\mathcal{P}$, a source-rendered image at frame t from viewpoint c_i is defined as

$$I_{t,i}^{\text{src}} = \mathcal{P}(M_t^{\text{src}}, c_i). \quad (9)$$

The corresponding source-rendered video is defined as $V_i^{\text{src}} = \{I_{t,i}^{\text{src}}\}_{t=0}^{T-1}$. For the reference viewpoint c_0 , this yields the reference video V_0^{src} used for multi-view synthesis. In addition, the first source mesh M_0^{src} is rendered from each target viewpoint, resulting in the target-view conditioning images $\{I_{0,i}^{\text{src}}\}_{i=1}^N$.

Subsequently, the synthesized target-view videos $\{\widehat{V}_i\}_{i=1}^N$ are reconstructed together with the source-rendered reference video V_0^{src} . For each source sequence, a corresponding ground-truth mesh sequence $\{M_t^{\text{gt}}\}_{t=0}^{T-1}$ is available and serves as the reference for evaluation of the reconstructed sequence $\{\widehat{M}_t\}_{t=0}^{T-1}$.

3.2.2 Dataset and Rendering

For the experiments, we use the CAPE dataset [29, 30], which provides sequences of clothed human meshes in motion together with corresponding SMPL body representations. The clothed meshes are used as the source sequences $\{M_t^{\text{src}}\}_{t=0}^{T-1}$ from which the source-rendered videos and target-view conditioning images are rendered.

For the evaluation of reconstruction accuracy, the corresponding unclothed SMPL meshes are used as ground-truth sequences $\{M_t^{\text{gt}}\}_{t=0}^{T-1}$. This distinction is important because the source renderings retain surface geometry such as clothing, whereas the reconstruction stage estimates the underlying SMPL body representation. Using clothed meshes as input therefore approximates a setting in which the observed surface differs from the underlying SMPL representation, as would also be the case when using an SMPL model as a deformation proxy for an arbitrary human mesh.

The source meshes are loaded using Trimesh and rendered using Pyrender. The camera is placed at a set of viewpoints c_i , where each viewpoint is parameterized only by its azimuth angle and the reference viewpoint is fixed at $c_0 = 0^\circ$. All remaining camera parameters are kept constant across the viewpoints of a sequence. The camera configuration is determined from the first source mesh M_0^{src} and then kept fixed over all frames.

The camera distance is set to 1.45 times the diagonal of the mesh bounding box. Its vertical position is offset by 0.25 times the bounding-box diagonal, while the viewing target is shifted upward by 0.05 times the same quantity. A perspective camera with a vertical field of view of 45° is used. The meshes are rendered with a uniform gray material on a white background.

3.2.3 Multi-View Video Synthesis

The diffusion model $\mathcal{G}$ of the multi-view synthesis stage is instantiated using SV4D 2.0 [8]. We select SV4D 2.0 because it provides publicly available weights and closely matches the required input and output modalities of our pipeline. While originally designed for 4D reconstruction, the model directly produces synchronized novel-view videos from an object-centric reference video, which can subsequently be used as input to our multi-view reconstruction stage. Internally, it improves spatio-temporal consistency by blending 3D attention with frame attention within its diffusion architecture. In our experiments, we use the four-view configuration and therefore set $N = 4$. Due to the available computational resources, inference is performed at a resolution of 512×512 pixels using 50 diffusion sampling steps.

The employed SV4D 2.0 checkpoint operates on generation windows of 12 frames. For longer sequences, its native autoregressive procedure uses frame-level reconditioning, where synthesized views from the preceding window are used to condition the subsequent generation window. In addition to this default strategy, we evaluate the mesh-level reconditioning approach introduced in Section 3.1.4. Its concrete implementation for long-horizon generation is described in Section 3.2.5.

For each source sequence, the source-rendered reference video V_0^{src} provides the temporal motion conditioning. The desired target viewpoints are specified by $\{c_i\}_{i=1}^N$, while the corresponding source-rendered images $\{I_{0,i}^{\text{src}}\}_{i=1}^N$ provide view-specific appearance conditioning.

To incorporate this target-view information, we modify the released SV4D 2.0 sampling procedure. In the original procedure, target-view conditioning is not available for the initial temporal window and is instead obtained from synthesized views for subsequent windows. In our implementation, the source-rendered target-view images $I_{0,i}^{\text{src}}$ are used to initialize this conditioning already for the first window.

We additionally disable the foreground cropping and rescaling performed by the original preprocessing pipeline. Preserving the original image geometry ensures that the known rendering configuration remains consistent with the synthesized target views and allows the corresponding camera parameters $\{\kappa_i\}_{i=0}^N$ to be used during subsequent multi-view reconstruction.

3.2.4 Multi-View Reconstruction

The reconstruction operator $\mathcal{R}$ is instantiated using an EasyMocap-based multi-view reconstruction pipeline [31]. For each sequence, the source-rendered reference video V_0^{src} and the synthesized target-view videos $\{\widehat{V}_i\}_{i=1}^N$ are provided together with their corresponding camera parameters $\{\kappa_i\}_{i=0}^N$. With $N = 4$, the reconstruction therefore operates on five synchronized views.

The required camera parameters are derived from the known rendering setup. Camera intrinsics are computed from the image resolution and the fixed vertical field of view, while the extrinsics follow from the corresponding camera positions and viewing directions. These parameters are provided to the reconstruction pipeline for both the source-rendered reference view and the synthesized target views.

For each frame, YOLOv5m [32] is first applied independently to all available camera views to obtain person bounding boxes. Within these regions, 2D body keypoints are estimated using HRNet-W48 [33]. Given the calibrated camera parameters, corresponding 2D keypoints across the synchronized views are then iteratively triangulated to obtain a sequence of 3D keypoints.

A neutral SMPL body model is subsequently fitted to the triangulated keypoints in multiple optimization stages. First, the body shape parameters are estimated with additional shape regularization. Global translation and orientation are then initialized from the triangulated 3D keypoints. Finally, body pose, global orientation, and translation are jointly refined using a 3D keypoint alignment objective together with temporal smoothness and a pose prior. The resulting SMPL parameters define the reconstructed mesh sequence $\{\widehat{M}_t\}_{t=0}^{T-1}$ used for subsequent evaluation.

The concrete EasyMocap configurations used in these stages are reported in Appendix A.2.

3.2.5 Long-Horizon Generation

The employed SV4D 2.0 checkpoint has a native generation window of 12 frames. After generating an initial window, SV4D 2.0 continues the synthesis autoregressively by using the final synthesized target-view frames as conditioning for the subsequent window. Consecutive full generation windows therefore overlap by one frame and advance by 11 frames. When generation starts at frame 0, the first full window covers frames 0–11, the second frames 11–22, and so forth. The final generation window before a desired temporal boundary may contain fewer than 12 frames. This corresponds to the frame-level autoregressive reconditioning strategy introduced in Section 3.1.4.

In the experimental evaluation, specific temporal points are selected at which frame-level and mesh-level reconditioning are compared. The generation-window partition is adjusted such that these reconditioning points coincide

with temporal boundaries τ_s , shortening the preceding generation window where necessary. At all remaining generation-window boundaries, the standard frame-level autoregressive continuation of SV4D 2.0 is retained. The specific reconditioning points used in the experiments are defined in Section 4.3.

For both reconditioning strategies, the generation window following a selected boundary is initialized through the modified initial target-view conditioning procedure described in Section 3.2.3. The corresponding subsequent temporal window of the source-rendered reference video $V_0^{\text{src},(s)}$ is used as the reference-video input in both cases. Thus, only the target-view conditioning at the selected boundary differs between the two strategies.

For frame-level reconditioning, the synthesized target-view frames $\{\widehat{I}_{\tau_s,i}^{(s-1)}\}_{i=1}^N$ at the shared boundary are directly forwarded as conditioning for the subsequent generation window, thereby reproducing the native frame-level continuation of SV4D 2.0. Due to GPU memory constraints, the individual generation windows are processed in separate inference passes, such that the boundary frames are explicitly transferred between consecutive passes.

For mesh-level reconditioning, the synthesized target-view videos $\{\widehat{V}_i^{(s-1)}\}_{i=1}^N$ of the preceding generation window are reconstructed together with the corresponding source-rendered reference window $V_0^{\text{src},(s-1)}$ using the EasyMocap-based reconstruction pipeline. The reconstructed mesh $\widehat{M}_{\tau_s}^{(s-1)}$ at the shared boundary is rendered from the same target viewpoints c_i , yielding the conditioning images $\{\mathcal{P}(\widehat{M}_{\tau_s}^{(s-1)}, c_i)\}_{i=1}^N$. These renderings replace the synthesized boundary frames as target-view conditioning for the subsequent generation window.

4 Experiments

This chapter evaluates the multi-view synthesis and reconstruction stages of the proposed human mesh animation architecture along four research questions. The evaluation is divided into two parts: the first analyzes the core multi-view setting, while the second investigates the additional challenges introduced by long-horizon generation.

Part I focuses on the general evaluation of the experimental multi-view pipeline. Its objective is to determine whether synthesized target-view videos provide useful additional constraints for recovering 3D human motion and to analyze how different pipeline configurations and components affect reconstruction performance. First, RQ1 evaluates whether synthesized target-view videos improve 3D motion reconstruction compared with single-view baselines (Section 4.2.1). RQ2 then investigates the role of the viewpoint configuration by analyzing both the angular spacing (Section 4.2.2) and the number of synthesized target views (Section 4.2.3). Finally, RQ3 examines the contribution of target-view conditioning using source-rendered images by

evaluating its effect on both synthesized target-view quality and downstream 3D reconstruction (Section 4.2.4).

Part II focuses on the evaluation of long-horizon motion generation. While the sequences evaluated in Part I are limited to comparatively short motion segments and use a fixed frame-level autoregressive reconditioning procedure, Part II considers substantially longer sequences and explicitly evaluates alternative reconditioning strategies. RQ4 therefore investigates how the different reconditioning strategies introduced in Section 3.1.4 affect the stability and quality of the synthesized target-view videos and the accuracy of the resulting downstream 3D reconstruction (Section 4.3.1).

4.1 Evaluation Protocol

The evaluation considers two complementary aspects of the proposed pipeline: 3D reconstruction accuracy and synthesized target-view quality. The following sections summarize the metrics and reporting procedure used throughout the experiments. Further mathematical definitions and implementation details are provided in Appendix A.3.

4.1.1 3D Reconstruction Metrics

Reconstruction accuracy is evaluated using complementary vertex- and joint-based metrics under different alignment conventions. MPVPE measures the average Euclidean distance between corresponding reconstructed and ground-truth mesh vertices. Before computing MPVPE, a single constant translation is estimated over the complete sequence and applied to all frames. This removes a sequence-wide translational offset that is irrelevant for the considered animation setting, while preserving errors in the relative global motion trajectory, orientation, and scale.

Joint-based accuracy is evaluated using pelvis-aligned MPJPE, where joint positions are obtained from the reconstructed and ground-truth meshes using the SMPL joint regressor. The reconstructed joints are translated independently in each frame such that the predicted pelvis coincides with the ground-truth pelvis. This removes global translation while retaining differences in orientation, scale, and articulated pose.

For both vertex- and joint-based errors, we additionally report Procrustes-aligned variants, PA-MPVPE and PA-MPJPE. A similarity transformation is estimated independently for each frame, removing translation, rotation, and uniform scale. These metrics therefore emphasize local mesh and pose accuracy rather than global placement.

Temporal consistency is evaluated using acceleration error, which compares the discrete second-order temporal differences of pelvis-aligned reconstructed and ground-truth joints. Lower acceleration error indicates closer agreement with the temporal evolution of the ground-truth motion.

All spatial reconstruction errors are reported in millimeters, while acceleration error is reported in mm/frame².

4.1.2 Synthesized Target-View Quality Metrics

Synthesized target-view quality is evaluated against the corresponding source-rendered target-view videos using PSNR, SSIM, and LPIPS. PSNR measures pixel-level fidelity and is reported in decibels (dB), with higher values indicating better agreement. SSIM measures structural image similarity, where higher values likewise indicate better quality. LPIPS measures perceptual similarity using features extracted by a pretrained neural network; in our implementation, an AlexNet-based feature extractor is used. Lower LPIPS values indicate greater perceptual similarity.

4.1.3 Reporting

Reconstruction metrics are first averaged over the matched frames of each motion sequence. The resulting sequence-level scores are then summarized across the evaluated sequences. Synthesized target-view quality metrics are computed for corresponding frame pairs, averaged over each target-view video, and subsequently summarized across all evaluated target-view videos of the respective configuration. Quantitative results are reported as mean $\pm$ sample standard deviation, computed across the resulting sequence-level or video-level scores, respectively.

Failed reconstruction evaluations are excluded from the corresponding aggregate statistics; any resulting deviation in the number of evaluated sequences is explicitly reported in the respective experiment.

4.2 Multi-View Reconstruction Evaluation

The first part of the evaluation investigates the reconstruction performance of the proposed multi-view pipeline and the influence of different synthesis and viewpoint configurations.

All experiments in this section use the same set of 33 CAPE sequences, selected at random from 7 subjects. The clothed mesh sequences $\{M_t^{\text{src}}\}_{t=0}^{T-1}$ are used to generate the source-rendered inputs, while the corresponding unclothed SMPL sequences $\{M_t^{\text{gt}}\}_{t=0}^{T-1}$ serve as ground truth for reconstruction evaluation.

The sequences are downsampled from 60 fps to 20 fps, after which the first 36 frames are retained for evaluation. Since this exceeds the native 12-frame generation window of SV4D 2.0, the model’s standard frame-level autoregressive continuation is used to extend the synthesis across consecutive generation windows. This native SV4D 2.0 procedure is kept fixed throughout RQ1–RQ3.

4.2.1 RQ1: Reconstruction Accuracy

The motivation for introducing synthesized target-view videos is to reduce the geometric ambiguity inherent in monocular 3D reconstruction. A single reference viewpoint exposes only a partial projection of the 3D information contained in the source mesh sequence due to missing depth information and self-occlusion. By using renderings from multiple target viewpoints to guide the synthesis of additional target-view videos, a larger portion of the available geometric information can be incorporated into the subsequent reconstruction. We therefore hypothesize that incorporating synthesized target-view videos improves 3D reconstruction accuracy compared with single-view reconstruction methods.

Experimental Setup. To evaluate this hypothesis, the proposed multi-view pipeline is compared against the two single-view human mesh reconstruction methods WHAM [34] and Multi-HMR [35]. WHAM performs video-based reconstruction and exploits temporal information across consecutive frames, whereas Multi-HMR reconstructs the human mesh independently for individual frames. The two methods therefore provide complementary single-view baselines.

For the proposed pipeline, $N = 4$ target-view videos are synthesized at the azimuth angles $\{-60^\circ, -30^\circ, 30^\circ, 60^\circ\}$ relative to the reference viewpoint $c_0 = 0^\circ$. The source-rendered reference video V_0^{src} and the synthesized target-view videos $\{\widehat{V}_i\}_{i=1}^4$ are subsequently reconstructed jointly using the EasyMocap-based multi-view reconstruction pipeline.

In addition, a source-rendered oracle is evaluated to isolate the effect of errors introduced by multi-view synthesis. Instead of the synthesized target-view videos $\{\widehat{V}_i\}_{i=1}^4$, the reconstruction is provided with the corresponding source-rendered videos $\{V_i^{\text{src}}\}_{i=1}^4$. The oracle reconstruction is therefore given by

$$\{\widehat{M}_t^{\text{oracle}}\}_{t=0}^{T-1} = \mathcal{R}((V_0^{\text{src}}, \kappa_0), \{(V_i^{\text{src}}, \kappa_i)\}_{i=1}^4). \quad (10)$$

The reference video, camera parameters, and reconstruction procedure are otherwise unchanged. This configuration provides a reference for the reconstruction accuracy attainable when the target-view observations are obtained directly from the source mesh sequence rather than synthesized by the MV-VDM.

Results. The proposed multi-view pipeline achieves the lowest MPVPE among the non-oracle methods (Table 1). Its MPVPE of 58.55 mm is substantially lower than the 112.57 mm obtained by WHAM and the 83.98 mm obtained by Multi-HMR. A similar improvement is observed for pelvis-aligned MPJPE, for which the proposed multi-view pipeline also outperforms both single-view baselines.

Table 1: 3D reconstruction accuracy of the evaluated methods.

Method	MPVPE	PA-MPVPE	Pelvis-Aligned		Acceleration
	↓ [mm]	↓ [mm]	MPJPE	PA-MPJPE	Error
			↓ [mm]	↓ [mm]	↓ [mm/f ²]
<i>Single-View Baselines</i>					
WHAM	112.57 ± 30.33	47.41 ± 9.09	66.03 ± 11.05	38.25 ± 6.46	11.98 ± 6.30
Multi-HMR	83.98 ± 14.52	41.06 ± 8.06	64.11 ± 9.97	32.83 ± 6.08	17.08 ± 7.34
<i>Synthesized Multi-Views</i>					
SV4D 2.0 + EasyMocap	58.55 ± 12.67	44.72 ± 9.76	49.55 ± 8.76	36.37 ± 6.58	10.48 ± 4.86
<i>Source-Rendered Oracle</i>					
EasyMocap	42.32 ± 9.15	35.70 ± 6.11	38.68 ± 7.81	28.89 ± 4.04	8.68 ± 4.07

In contrast, Multi-HMR performs best among the non-oracle methods on the Procrustes-aligned metrics. For PA-MPVPE, Multi-HMR achieves 41.06 mm compared with 44.72 mm for the proposed multi-view pipeline. At the same time, Multi-HMR exhibits the highest acceleration error among the compared methods, whereas the proposed multi-view pipeline achieves the lowest acceleration error among the non-oracle methods.

The source-rendered oracle achieves the lowest reconstruction error across all evaluated metrics. In particular, its MPVPE decreases to 42.32 mm compared with 58.55 mm when synthesized target views are used.

Qualitative inspection of the reconstructed sequences shows a similar pattern (Figure 2). WHAM exhibits noticeable global drift, while Multi-HMR produces locally accurate poses but stronger frame-to-frame jitter. In comparison, the proposed pipeline maintains a more stable global position while preserving competitive local pose accuracy. The source-rendered oracle shows the closest qualitative agreement with the corresponding ground-truth motion.

Overall, the results largely support the hypothesis. The additional synthesized target views substantially improve global reconstruction accuracy compared with the evaluated single-view baselines. Although Multi-HMR achieves better locally aligned pose accuracy, its higher acceleration error and visible frame-to-frame jitter indicate lower temporal consistency, which is consistent with the frame-wise nature of the method. We argue that for the target use case of mesh animation, the proposed multi-view pipeline therefore provides a more favorable balance between global motion accuracy and temporal stability. The performance gap to the source-rendered oracle further indicates that inaccuracies in the synthesized target views remain a major limiting factor for downstream multi-view reconstruction.

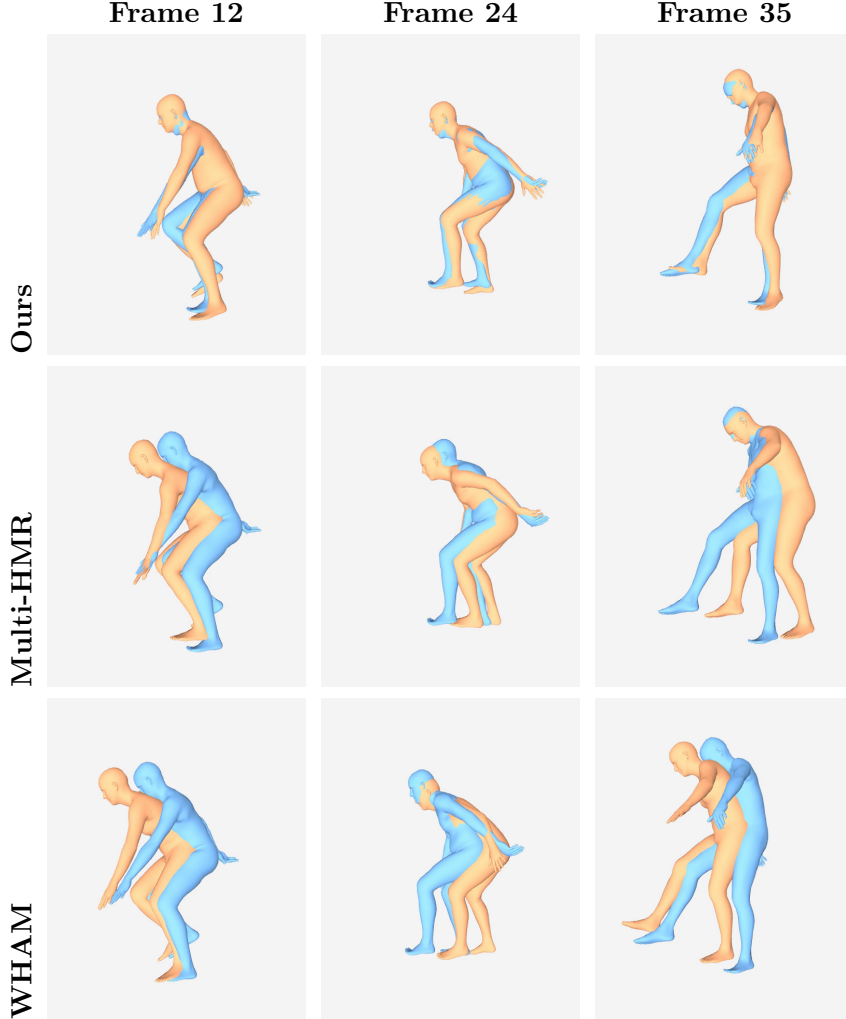


Figure 2: Qualitative comparison of reconstruction results for the proposed multi-view pipeline (*Ours*) and the single-view baselines Multi-HMR and WHAM across three temporally separated frames of the same motion sequence. Each panel shows an overlay of the reconstructed mesh (orange) and the corresponding ground-truth mesh (blue).

4.2.2 RQ2a: Viewpoint Configuration – Angular Spacing

We expect the angular spacing between viewpoints to introduce two competing effects. On the one hand, larger viewpoint differences provide broader geometric coverage and may therefore impose stronger constraints on multi-view reconstruction. On the other hand, synthesizing target views farther from the reference viewpoint may be more challenging for the multi-view VDM and reduce the fidelity of the generated videos. RQ2a therefore investigates whether and to what extent this trade-off occurs in practice and

which angular spacing yields the best overall reconstruction performance for the proposed pipeline.

Experimental Setup. We compare three five-view configurations parameterized by the angular spacing $x \in \{30^\circ, 45^\circ, 72^\circ\}$. For a given spacing x , the viewpoint configuration is defined as $\mathcal{Q}_x = \{-2x, -x, 0^\circ, x, 2x\}$, where the reference viewpoint is fixed at $c_0 = 0^\circ$ and the remaining four angles correspond to the target viewpoints $\{c_i\}_{i=1}^4$. The number of synthesized target views is therefore kept fixed at $N = 4$, and only their angular placement is varied.

The evaluation is performed at three stages of the pipeline. First, the source-rendered oracle procedure introduced in Section 4.2.1 is applied to each viewpoint configuration. Since all target-view observations are rendered directly from the source mesh sequence, this comparison isolates the effect of angular spacing on the multi-view reconstruction itself. Second, the fidelity of the synthesized target-view videos $\{\widehat{V}_i\}_{i=1}^4$ is evaluated by comparing them with the corresponding source-rendered target-view videos $\{V_i^{\text{src}}\}_{i=1}^4$. Finally, the synthesized target views are used together with the source-rendered reference video V_0^{src} for multi-view reconstruction.

This three-stage evaluation allows the effect of geometric viewpoint coverage to be considered separately from the loss in fidelity introduced by multi-view synthesis. For the synthesized-view reconstruction at 72° , one sequence could not be successfully reconstructed and is therefore excluded from the quantitative reconstruction evaluation.

Results. When reconstruction is performed entirely from source-rendered views, increasing the angular spacing consistently reduces the reconstruction error across the evaluated metrics (Table 2). The 72° configuration achieves the lowest reconstruction errors, although the differences between the three configurations are comparatively small.

Table 2: 3D reconstruction accuracy using source-rendered views at different angular spacings.

Angular Spacing	MPVPE ↓ [mm]	PA-MPVPE ↓ [mm]	Pelvis-Aligned	
			MPJPE ↓ [mm]	PA-MPJPE ↓ [mm]
30°	42.32 ± 9.15	35.70 ± 6.11	38.68 ± 7.81	28.89 ± 4.04
45°	41.54 ± 8.94	34.83 ± 6.28	37.32 ± 7.69	27.84 ± 4.16
72°	40.69 ± 9.29	34.31 ± 6.43	36.99 ± 8.02	27.41 ± 4.24

The opposite trend is observed for synthesized-view fidelity (Table 3). The 30° configuration achieves the best PSNR, SSIM, and LPIPS scores, while fidelity decreases as the angular spacing increases. The 72° configu-

ration performs worst across all three metrics, with the largest degradation occurring between the 45° and 72° configurations.

Table 3: Synthesized target-view quality at different angular spacings.

Angular Spacing	PSNR ↑ [dB]	SSIM ↑	LPIPS ↓
30°	27.43 ± 1.41	0.9491 ± 0.0089	0.0524 ± 0.0124
45°	26.94 ± 1.40	0.9478 ± 0.0097	0.0591 ± 0.0142
72°	25.92 ± 1.56	0.9414 ± 0.0099	0.0730 ± 0.0203

The downstream reconstruction results obtained from synthesized target views reflect this decrease in synthesis fidelity (Table 4). Depending on the reconstruction metric, either the 30° or 45° configuration performs best, whereas the 72° configuration performs worst across all evaluated reconstruction metrics. The difference is most pronounced for MPVPE, which increases from 58.55 mm for the 30° configuration to 68.65 mm for the 72° configuration.

Table 4: 3D reconstruction accuracy using synthesized target views at different angular spacings. For the 72° configuration, one failed reconstruction is excluded from the evaluation.

Angular Spacing	MPVPE ↓ [mm]	PA-MPVPE ↓ [mm]	Pelvis-Aligned	
			MPJPE ↓ [mm]	PA-MPJPE ↓ [mm]
30°	58.55 ± 12.67	44.72 ± 9.76	49.55 ± 8.76	36.37 ± 6.58
45°	59.11 ± 13.28	45.16 ± 10.93	48.45 ± 9.75	36.31 ± 7.25
72°	68.65 ± 18.87	46.55 ± 10.84	54.27 ± 12.76	36.99 ± 7.64

Overall, the results confirm the expected trade-off between geometric coverage and synthesized-view fidelity. When synthesis errors are removed through the source-rendered oracle, wider angular spacing provides a small but consistent benefit for multi-view reconstruction. In the complete pipeline, however, this benefit is outweighed by the reduced fidelity of target views synthesized at larger angular offsets. Moderate angular spacings are therefore more favorable for the proposed pipeline. Since the 30° and 45° configurations achieve the best performance on different reconstruction metrics, the experiments do not identify a single optimal angular spacing.

4.2.3 RQ2b: Viewpoint Configuration – Number of Views

We expect reconstruction accuracy to improve as more viewpoints are included, since additional target views provide complementary observations of the motion and may reduce ambiguities in the multi-view reconstruction. At the same time, each additional synthesized view can introduce noise or

cross-view inconsistencies that may limit the benefit of increasing the number of views. RQ2b therefore investigates whether reconstruction accuracy improves with a larger number of views and whether this improvement depends on the angular placement of the selected viewpoints.

Experimental Setup. The experiment is based on the five-view configuration $\{-60^\circ, -30^\circ, 0^\circ, 30^\circ, 60^\circ\}$ used in RQ1. Since the employed SV4D 2.0 configuration generates $N = 4$ target views in addition to the reference view, this represents the maximum number of views considered in this experiment. To evaluate the influence of view count, subsets containing between three and five total views are provided to the reconstruction pipeline, while always retaining the reference viewpoint $c_0 = 0^\circ$.

A two-view configuration is not considered because the employed Easy-Mocap pipeline relies on iterative multi-view triangulation and requires at least three valid observations for the reconstruction of a 3D keypoint. To account for the fact that reconstruction performance may depend not only on the number of views but also on their angular placement, two different three-view configurations are evaluated. We define $\mathcal{Q}_{3a} = \{-30^\circ, 0^\circ, 30^\circ\}$, $\mathcal{Q}_{3b} = \{-60^\circ, 0^\circ, 30^\circ\}$, $\mathcal{Q}_4 = \{-60^\circ, 0^\circ, 30^\circ, 60^\circ\}$, and $\mathcal{Q}_5 = \{-60^\circ, -30^\circ, 0^\circ, 30^\circ, 60^\circ\}$.

Results. Using all five views in $\mathcal{Q}_5$ achieves the lowest reconstruction error across all evaluated metrics (Table 5). However, the improvement over configurations with fewer views is comparatively small. For MPVPE, $\mathcal{Q}_5$ achieves 58.55 mm, compared with 60.02 mm for $\mathcal{Q}_{3b}$ and 61.01 mm for $\mathcal{Q}_4$.

Table 5: 3D reconstruction accuracy for different viewpoint configurations.

Config.	Viewpoints	MPVPE ↓ [mm]	PA-MPVPE ↓ [mm]	Pelvis-Aligned	
				MPJPE ↓ [mm]	PA-MPJPE ↓ [mm]
$\mathcal{Q}_{3a}$	$\{0^\circ, \pm 30^\circ\}$	63.92 ± 16.13	47.44 ± 11.49	53.80 ± 11.30	38.57 ± 7.64
$\mathcal{Q}_{3b}$	$\{-60^\circ, 0^\circ, 30^\circ\}$	60.02 ± 13.96	46.31 ± 10.38	50.65 ± 10.45	37.46 ± 7.22
$\mathcal{Q}_4$	$\{0^\circ, 30^\circ, \pm 60^\circ\}$	61.01 ± 13.86	46.10 ± 10.68	51.02 ± 9.29	37.47 ± 7.20
$\mathcal{Q}_5$	$\{0^\circ, \pm 30^\circ, \pm 60^\circ\}$	58.55 ± 12.67	44.72 ± 9.76	49.55 ± 8.76	36.37 ± 6.58

The results are not monotonic with respect to the number of views. In particular, $\mathcal{Q}_{3b}$ slightly outperforms $\mathcal{Q}_4$ on three of the four evaluated reconstruction metrics, despite using one fewer view. The comparison between the two three-view configurations further shows that reconstruction performance also depends on angular placement: $\mathcal{Q}_{3b}$ consistently outperforms $\mathcal{Q}_{3a}$ across all evaluated metrics.

Overall, the results partially support the expectation that additional views improve reconstruction accuracy. While $\mathcal{Q}_5$ provides the best overall performance, increasing the number of views does not lead to a monotonic reduction in reconstruction error. The differences between $\mathcal{Q}_{3a}$ and $\mathcal{Q}_{3b}$ further

indicate that the usefulness of additional views depends not only on their number but also on their angular placement. Consequently, the experiment does not establish that adding further synthesized views beyond the evaluated five-view configuration would necessarily provide additional improvements. Future work could extend this analysis to larger view counts, for example using an eight-view configuration of SV4D 2.0, to investigate whether additional synthesized viewpoints provide further reconstruction benefits or whether the observed gains saturate.

4.2.4 RQ3: Target-View Conditioning

Since the source mesh is known, renderings from the target viewpoints can provide additional view-specific appearance information during multi-view synthesis. We hypothesize that this target-view conditioning reduces visual drift and improves the fidelity of the synthesized target-view videos. We further expect that the resulting improvement in synthesis fidelity translates into higher downstream 3D reconstruction accuracy compared with multi-view synthesis without target-view conditioning.

Experimental Setup. All experiments use the same five-view configuration as in RQ1, with the reference viewpoint fixed at $c_0 = 0^\circ$ and the four target viewpoints at azimuth angles $\{-60^\circ, -30^\circ, 30^\circ, 60^\circ\}$. Target-view conditioning is the only varied component.

For the conditioned configuration, the source-rendered target-view images $\{I_{0,i}^{\text{src}}\}_{i=1}^4$ are used as target-view conditioning, as defined in Section 3.2.3, yielding the synthesized target-view videos $\{\widehat{V}_i\}_{i=1}^4$. For the unconditioned configuration, the target-view conditioning is omitted. The corresponding synthesized target-view videos are defined as

$$\{\widehat{V}_i^{\text{uncond}}\}_{i=1}^4 = \mathcal{G}(V_0^{\text{src}}, \{\mathbf{0}\}_{i=1}^4, \{c_i\}_{i=1}^4), \quad (11)$$

where $\mathbf{0}$ denotes the absence of target-view conditioning. All remaining inputs and generation parameters are kept unchanged.

The resulting conditioned and unconditioned target-view videos are evaluated in two steps. First, their synthesis fidelity is measured by comparing $\{\widehat{V}_i\}_{i=1}^4$ and $\{\widehat{V}_i^{\text{uncond}}\}_{i=1}^4$, respectively, with the corresponding source-rendered target-view videos $\{V_i^{\text{src}}\}_{i=1}^4$. Second, the conditioned and unconditioned target-view videos are each used, together with the same source-rendered reference video V_0^{src} , as input to the multi-view reconstruction stage. This allows the effect of target-view conditioning on synthesis fidelity to be compared with its effect on downstream reconstruction accuracy.

Results. Target-view conditioning substantially improves synthesized-view fidelity across all evaluated metrics (Table 6). PSNR increases from 21.22 dB

without conditioning to 27.43 dB with conditioning, while LPIPS decreases from 0.1078 to 0.0524. SSIM shows the same trend, increasing from 0.9172 to 0.9491. Qualitatively, the conditioned target-view videos remain more consistent with the appearance of the source mesh over time, whereas the unconditioned generation exhibits stronger and earlier visual drift (Figure 3).

Table 6: Synthesized target-view quality with and without source-rendered target-view conditioning.

Generation Setting	PSNR ↑ [dB]	SSIM ↑	LPIPS ↓
Conditioned	27.43 ± 1.41	0.9491 ± 0.0089	0.0524 ± 0.0124
Unconditioned	21.22 ± 1.33	0.9172 ± 0.0082	0.1078 ± 0.0248

(a) Source-Rendered



(b) Unconditioned



(c) Conditioned

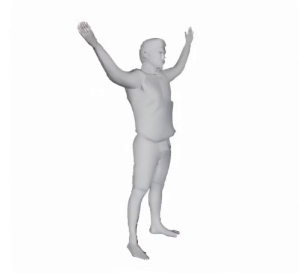


Figure 3: Qualitative comparison of synthesized target views with and without source-rendered target-view conditioning. All three images show the same target view-point and the same frame of the same motion sequence.

The improvement in synthesized-view fidelity is also reflected in downstream reconstruction accuracy (Table 7). MPVPE decreases from 66.68 mm without target-view conditioning to 58.55 mm with conditioning. Pelvis-aligned MPJPE similarly improves, while the difference is less pronounced for the Procrustes-aligned pose metric. Compared with the single-view baseline Multi-HMR, the unconditioned multi-view configuration therefore retains only part of the reconstruction improvement achieved by the conditioned pipeline.

Table 7: 3D reconstruction accuracy with and without source-rendered target-view conditioning compared with single-view baseline.

Method	MPVPE	PA-MPVPE	Pelvis-Aligned MPJPE	PA-MPJPE
	↓ [mm]	↓ [mm]	↓ [mm]	↓ [mm]
<i>Single-View Baseline</i>				
Multi-HMR	83.98 ± 14.52	41.06 ± 8.06	64.11 ± 9.97	32.83 ± 6.08
<i>SV4D 2.0 + EasyMocap</i>				
Unconditioned	66.68 ± 10.59	47.00 ± 8.11	58.12 ± 10.36	38.26 ± 5.64
Conditioned	58.55 ± 12.67	44.72 ± 9.76	49.55 ± 8.76	36.37 ± 6.58

Overall, the results support the hypothesis that target-view conditioning benefits both multi-view synthesis and downstream reconstruction. The particularly large improvement in synthesized-view fidelity shows that exploiting view-specific information from the known source mesh substantially stabilizes the generated target views. These improvements also reduce reconstruction error, although the gain in reconstruction accuracy is less pronounced than the improvement in image-based fidelity metrics. This indicates that improvements in visual similarity do not translate proportionally into improvements in 3D reconstruction accuracy. Since this conditioning strategy requires access to the input 3D mesh, it is inherently limited to animation settings in which the input mesh is known in advance, which is consistent with the intended use case of this work.

4.3 Long-Horizon Generation

The second part of the evaluation investigates the stability of the proposed multi-view pipeline over substantially longer generation horizons and analyzes the influence of the different reconditioning strategies introduced in Section 3.1.4. In contrast to the experiments in Section 4.2, this evaluation uses 5 CAPE sequences from 3 subjects and considers the first 106 frames of each sequence after downsampling from 60 fps to 20 fps.

For this experiment, the unclothed SMPL sequences provided by CAPE are used directly as source sequences $\{M_t^{\text{src}}\}_{t=0}^{T-1}$ and as ground-truth motion $\{M_t^{\text{gt}}\}_{t=0}^{T-1}$ for reconstruction evaluation. This differs from Part I, where clothed meshes are used as source sequences. As described in Section 3.2.4, the reconstruction stage estimates an SMPL body representation and therefore does not recover the clothing geometry present in the clothed CAPE meshes. Consequently, mesh-level reconditioning with clothed source sequences would introduce an additional appearance change at the reconditioning boundary, since the conditioning views rendered from the reconstructed mesh would be unclothed. Using unclothed source sequences allows the experiment to isolate the effect of the reconditioning strategy itself.

In a practical application to arbitrary human meshes, this limitation could be addressed using the motion-transfer procedure described in Section 3.1.5: the reconstructed SMPL motion can be transferred back to the original input mesh before rendering the conditioning views for subsequent generation.

4.3.1 RQ4: Reconditioning Strategies

Repeated frame-level autoregressive reconditioning over long generation horizons may accumulate errors in the synthesized target views and consequently degrade downstream reconstruction accuracy. We hypothesize that mesh-level reconditioning can reduce this error accumulation by projecting the generated motion back into an explicit 3D representation before conditioning subsequent generation. We therefore expect mesh-level reconditioning to preserve both synthesized-view fidelity and reconstruction accuracy better than frame-level reconditioning over increasing generation horizons.

Experimental Setup. To evaluate long-horizon stability, each sequence is extended to 106 frames, and the frame-level and mesh-level reconditioning strategies are compared at two selected reconditioning boundaries at frames 35 and 70. The generation-window partition is chosen such that these points coincide with shared temporal boundaries: $\tau_4 = 35, \tau_8 = 70$.

Where necessary, the generation window preceding a selected reconditioning boundary is shortened accordingly. At all other generation-window boundaries, the standard frame-level autoregressive continuation of SV4D 2.0 is retained. All runs therefore share the same generation up to frame 35 and differ only in the target-view conditioning applied at the two selected reconditioning boundaries.

The frame-level and mesh-level reconditioning strategies follow the definitions in Section 3.1.4. In addition, ground-truth reconditioning is included as an oracle reference. At both of the selected reconditioning boundaries τ_s , the corresponding ground-truth mesh $M_{\tau_s}^{\text{gt}}$ is rendered from the target viewpoints, yielding the conditioning images $\{\mathcal{P}(M_{\tau_s}^{\text{gt}}, c_i)\}_{i=1}^4$ for generation window s . This oracle reference represents long-horizon continuation with geometrically correct motion at the reconditioning boundary.

For all three variants, the corresponding source-rendered reference window $V_0^{\text{src},(s)}$ is used as the reference-video input for generation window s . The variants therefore differ only in the target-view conditioning applied at the selected reconditioning boundaries.

All variants are evaluated in two ways. First, synthesized-view fidelity is measured separately up to the first reconditioning boundary, between the two reconditioning boundaries, and after the second boundary to analyze how synthesis quality changes over increasing generation horizons. Second, reconstruction accuracy is evaluated over the complete 106-frame sequence.

In addition to the aggregate results over the full sequence, all reconstruction metrics are computed per frame to analyze their temporal development and, in particular, changes around the two selected reconditioning boundaries.

Results. Both autoregressive reconditioning strategies exhibit decreasing synthesized-view fidelity over increasing generation horizons, whereas ground-truth reconditioning remains comparatively stable (Table 8). After the first selected reconditioning boundary, mesh-level reconditioning achieves slightly higher fidelity than frame-level reconditioning. This difference becomes more pronounced after the second boundary. For frames 71–105, mesh-level reconditioning achieves an LPIPS of 0.0920 compared with 0.1014 for frame-level reconditioning, while ground-truth reconditioning reaches 0.0547. PSNR and SSIM show the same overall ordering.

Table 8: Synthesized target-view quality under different reconditioning strategies across increasing generation horizons.

Frame Range	Reconditioning Strategy	PSNR ↑ [dB]	SSIM ↑	LPIPS ↓
0–35	Initial Conditioning	27.26 ± 0.84	0.9494 ± 0.0071	0.0513 ± 0.0085
36–70	Ground Truth	26.64 ± 1.01	0.9473 ± 0.0060	0.0580 ± 0.0121
36–70	Mesh-Level	24.46 ± 1.14	0.9348 ± 0.0085	0.0835 ± 0.0182
36–70	Frame-Level	23.72 ± 1.43	0.9330 ± 0.0078	0.0847 ± 0.0189
71–105	Ground Truth	26.70 ± 1.26	0.9483 ± 0.0090	0.0547 ± 0.0117
71–105	Mesh-Level	23.96 ± 1.20	0.9337 ± 0.0113	0.0920 ± 0.0200
71–105	Frame-Level	22.08 ± 1.52	0.9278 ± 0.0092	0.1014 ± 0.0196

The reconstruction results show a different trend (Table 9). Despite the higher synthesized-view fidelity obtained with mesh-level reconditioning, frame-level reconditioning achieves lower reconstruction error across all evaluated metrics. Over the complete 106-frame sequence, frame-level reconditioning achieves an MPVPE of 66.34 mm compared with 69.65 mm for mesh-level reconditioning. Ground-truth reconditioning provides the lowest reconstruction error, with an MPVPE of 56.55 mm. The remaining reconstruction metrics exhibit the same ordering.

Table 9: Overall 3D reconstruction accuracy under different reconditioning strategies.

Reconditioning Strategy	MPVPE ↓ [mm]	PA-MPVPE ↓ [mm]	Pelvis-Aligned	
			MPJPE ↓ [mm]	PA-MPJPE ↓ [mm]
Ground Truth	56.55 ± 8.05	43.66 ± 4.52	46.25 ± 5.67	34.60 ± 3.20
Frame-Level	66.34 ± 11.47	48.41 ± 6.07	50.66 ± 7.37	37.89 ± 4.87
Mesh-Level	69.65 ± 10.45	50.09 ± 6.81	52.90 ± 3.92	40.05 ± 4.33

The per-frame reconstruction errors further reveal distinct changes around the selected reconditioning boundaries (Figure 4). Ground-truth reconditioning produces a clear reduction in reconstruction error after each boundary, whereas frame-level reconditioning leads to a slight increase and mesh-level reconditioning to a more pronounced increase. A particularly strong rise in reconstruction error occurs for mesh-level reconditioning around frame 80. This spike is primarily attributable to a reconstruction failure in one of the evaluated sequences.

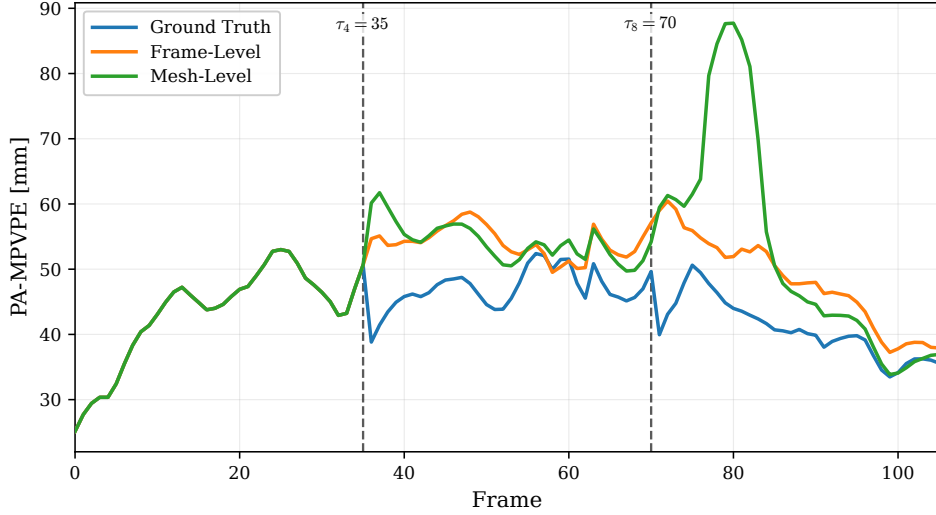


Figure 4: Per-frame PA-MPVPE for the three reconditioning strategies over the 106-frame long-horizon sequences. The curves show the mean across the five evaluated sequences. The dashed vertical lines indicate the selected reconditioning boundaries at frames 35 and 70.

Qualitative inspection of this failure case shows that one arm is reconstructed incorrectly. As synthesized-view fidelity decreases, the arm becomes increasingly difficult to distinguish from the torso in the corresponding target-view videos, reducing the useful geometric information available to the subsequent multi-view reconstruction.

Overall, the results only partially support the hypothesis. Neither frame-level nor mesh-level reconditioning remains fully stable over long generation horizons, and both show stronger degradation in synthesized-view fidelity and reconstruction accuracy over time than ground-truth reconditioning. While mesh-level reconditioning preserves synthesized-view fidelity better than frame-level reconditioning, this advantage does not translate into improved 3D reconstruction accuracy. Instead, frame-level reconditioning achieves lower reconstruction error across all evaluated metrics. A possible explanation lies in the different types of error propagated by the two strategies. Mesh-level reconditioning introduces the reconstruction stage into the

autoregressive feedback loop. Errors in the reconstructed pose or geometry are therefore encoded in the rendered conditioning images and may become inconsistent with the continuing source-rendered reference video. Frame-level reconditioning does not introduce such reconstruction errors, but instead directly propagates visual distortions from previously synthesized frames, which can accumulate over time. In the evaluated sequences, the geometric errors introduced by mesh-level reconditioning appear to have a stronger negative effect on downstream reconstruction than the visual distortions propagated by frame-level reconditioning.

The discrepancy between synthesized-view fidelity and reconstruction accuracy also highlights a limitation of the image-based fidelity metrics used in this work. PSNR, SSIM, and LPIPS measure visual similarity but do not directly capture multi-view geometric consistency. Consequently, higher synthesized-view fidelity does not necessarily imply that the synthesized target views provide more accurate geometric constraints for downstream reconstruction.

5 Conclusion and Future Work

5.1 Summary

This project investigated whether a pretrained general-purpose multi-view video diffusion model can be used as part of a human mesh animation pipeline by synthesizing additional target-view videos for multi-view 3D reconstruction. The experimental evaluation focused on the multi-view synthesis and reconstruction stages and examined the reconstruction benefit of synthesized target views, the influence of different pipeline configurations, and the continuation of multi-view generation beyond the model’s native generation window.

The results show that synthesized target-view videos can provide useful additional constraints for human motion reconstruction. Compared with the two evaluated single-view baselines, Multi-HMR and WHAM, the proposed multi-view pipeline reduces MPVPE to 58.55 mm, from 83.98 mm and 112.57 mm, respectively. At the same time, Multi-HMR performs better on the Procrustes-aligned metrics, indicating that the main benefit of the multi-view approach lies in reconstruction aspects that retain more information about global motion. The source-rendered oracle further shows that inaccuracies in the synthesized target views remain an important limiting factor.

The configuration experiments demonstrate that the benefit of additional views depends on both their placement and their synthesis quality. Increasing the angular spacing introduces two opposing effects: wider viewpoint coverage provides stronger geometric constraints for reconstruction, while larger angular offsets reduce the fidelity of the synthesized target views. In

the evaluated configurations, the loss in synthesis fidelity outweighs the geometric benefit at larger offsets, such that the moderate 30° and 45° spacings perform better than the 72° configuration in downstream reconstruction. Increasing the number of synthesized target views can also improve reconstruction performance, with the configuration using all five available views achieving the best overall results. However, the improvement is not monotonic and remains comparatively small, showing that simply adding more synthesized views does not necessarily lead to better reconstruction.

Target-view conditioning provides another clear benefit to the proposed pipeline. Supplying source-rendered images from the target viewpoints substantially improves the fidelity and temporal stability of the synthesized views, reducing LPIPS from 0.1078 to 0.0524, and also improves downstream reconstruction, with MPVPE decreasing from 66.68 mm to 58.55 mm. However, as reflected in these results, the improvement in synthesized-view fidelity is considerably stronger than the improvement in reconstruction accuracy, indicating that higher visual similarity does not translate proportionally into more accurate 3D reconstruction.

To extend multi-view generation beyond the model’s native generation window, two autoregressive reconditioning strategies were evaluated. Frame-level reconditioning reuses the synthesized target-view frames at a generation-window boundary to condition the following window, whereas mesh-level reconditioning first reconstructs the motion at the boundary and renders the reconstructed mesh from the target viewpoints for reconditioning. Both strategies show increasing degradation over longer generation horizons. Mesh-level reconditioning preserves synthesized-view fidelity better than frame-level reconditioning, achieving an LPIPS of 0.0920 compared with 0.1014 after the second reconditioning boundary. However, this advantage does not translate into better downstream reconstruction: frame-level reconditioning achieves a lower MPVPE of 66.34 mm than mesh-level reconditioning with 69.65 mm. Ground-truth reconditioning performs substantially better, indicating that accurate conditioning at generation boundaries is important for reliable long-horizon continuation.

Overall, the results demonstrate the potential of general-purpose multi-view video diffusion for human mesh animation, while also showing that reconstruction performance depends strongly on the viewpoint configuration and that reliable continuation over longer generation horizons remains a key challenge.

5.2 Limitations

The experimental evaluation is intentionally conducted in a controlled setting rather than as a complete end-to-end animation pipeline. In particular, the single-view video diffusion stage is replaced by source-rendered reference videos, allowing the multi-view synthesis and reconstruction stages to

be evaluated independently. Consequently, the experiments do not capture how errors introduced during single-view motion generation would propagate through the subsequent multi-view synthesis and reconstruction stages. Similarly, although the proposed architecture can conceptually be extended to arbitrary human meshes through registration and motion transfer, this extension is not implemented or evaluated in the present work.

A second limitation is the scale of the experimental evaluation. While the multi-view reconstruction experiments are conducted on 33 motion sequences, the long-horizon evaluation is restricted to five sequences from three subjects. Due to the computational cost of multi-view video diffusion, only a limited number and length of sequences can be evaluated within the available resources. This computational constraint also limits the experiments to four synthesized target views. Consequently, the results provide evidence for the investigated configurations, but do not constitute an exhaustive evaluation across larger datasets, longer motions, or higher view counts.

Finally, the employed multi-view reconstruction pipeline was not specifically designed for observations synthesized by a video diffusion model. The source-rendered reference video and synthesized target views are processed using the same detection, keypoint estimation, triangulation, and SMPL fitting procedure, although synthesized views may contain view-dependent artifacts and cross-view inconsistencies that are not present in geometrically consistent multi-view observations. While the iterative triangulation procedure provides some robustness to inconsistent observations, synthesis uncertainty is not modeled explicitly. Moreover, the reconstruction pipeline estimates the SMPL body shape as part of the fitting process, although the target body shape would already be known from the input human mesh in the intended application setting. The discrepancy observed between image-based fidelity and reconstruction accuracy further indicates that metrics such as PSNR, SSIM, and LPIPS do not directly capture the geometric consistency that is most relevant for multi-view reconstruction.

5.3 Future Work

A natural next step is to evaluate the proposed architecture as a complete end-to-end animation pipeline. This would require generating the initial reference video from a rendering of the input mesh and an action description instead of using a controlled source-rendered reference sequence. Such an evaluation would make it possible to study how errors introduced by the single-view video diffusion model propagate through multi-view synthesis and subsequent reconstruction. As an intermediate controlled experiment, source-rendered reference videos could be partially noised and subsequently denoised with a VDM, following an SDEdit-inspired procedure [36], to introduce controllable diffusion-model artifacts. Varying the denoising strength would allow the sensitivity of the subsequent pipeline stages to different lev-

els of reference-video degradation to be analyzed systematically. In addition, the registration and motion-transfer procedure could be implemented to extend the evaluation to arbitrary human meshes rather than only recovering motion in the SMPL representation.

A second direction concerns reconstruction from synthesized multi-view observations. A reconstruction method designed specifically for this setting could account explicitly for the different reliability of the reference and synthesized views, for example through confidence-based view weighting or more robust treatment of inconsistent observations during triangulation and fitting. In addition, the known target body shape could be incorporated directly into the reconstruction instead of being estimated again from the synthesized observations. More computationally expensive optimization-based objectives could also be investigated, including image-, silhouette-, or feature-based alignment inspired by existing video-based mesh animation approaches such as Animating the Uncaptured [7]. Since multi-view video synthesis currently dominates the computational cost of the pipeline, allocating additional computation to a more accurate reconstruction stage may be a worthwhile trade-off. Replacing individual detection, pose-estimation, and reconstruction components with more recent alternatives could provide further improvements.

Finally, the practical applicability of the proposed approach depends on reliable continuation beyond the native generation window. Although the experiments did not show an advantage of mesh-level reconditioning for downstream reconstruction, its effectiveness is likely to depend on the accuracy of the reconstructed mesh used for reconditioning. A reconstruction method that is more robust to synthesized observations could therefore improve both the final motion estimate and the reliability of mesh-level reconditioning. Future work could also investigate hybrid strategies that combine information from synthesized boundary frames with geometry obtained from reconstruction, rather than relying exclusively on either frame-level or mesh-level reconditioning. Such approaches may provide a better balance between preserving visual appearance and maintaining geometric consistency over longer generation horizons.

A Appendix: Additional Materials

A.1 Software Environments

Different processing stages of the experimental pipeline were executed in separate Conda environments due to their differing software requirements. Mesh rendering used Python 3.9.25 together with Trimesh 4.12.2 and Pyrender 0.1.45. SMPL-based evaluation was performed using Python 3.10.20, PyTorch 2.12.1, and SMPL-X 0.1.28. Synthesized-view quality evaluation used Python 3.14.6, PyTorch 2.10.0, TorchMetrics 1.9.0, and CUDA 12.9. Only

the main dependencies relevant to the corresponding processing stages are reported here; more detailed environment specifications are provided with the accompanying code.

A.2 EasyMocap Reconstruction Configuration

The reconstruction procedure is described in Section 3.2.4. This appendix reports the concrete EasyMocap configuration used throughout the experiments.

Detection and keypoint estimation. Person detection uses YOLOv5m restricted to the person class. 2D keypoints are estimated using HRNet-W48 with the `pose_hrnet_w48_384x288` checkpoint. Multi-view triangulation uses EasyMocap’s iterative triangulation mode.

SMPL configuration. The fitting procedure uses a neutral SMPL model (`SMPL_NEUTRAL.pkl`) together with the Body25 joint regressor. The model uses 69 pose parameters and 10 shape parameters. The Body25 regressor is used internally by the EasyMocap fitting procedure and is distinct from the 24-joint SMPL regressor used for the reconstruction metrics described in Appendix A.3. All fitting stages are optimized with L-BFGS.

Shape and global initialization. Shape parameters are optimized using a limb-length loss with weight 100 and an ℓ_2 shape regularization term with weight 0.1. Global rotation and translation are subsequently initialized using a 3D keypoint loss with weight 100 on keypoints [2, 5, 9, 12], together with second-order temporal smoothness.

Pose refinement. Body pose, global rotation, and translation are jointly refined in two optimization passes. The refinement objective combines a 3D keypoint loss with weight 1000, temporal smoothness with weight 1, and an eight-component GMM pose prior derived from SMPLify [37] with weight 0.1. The 3D keypoint loss uses `norm_info=0.02`.

Temporal smoothness. During global initialization, the relative smoothness weights for global translation and 3D keypoints are 100 and 10, respectively. During pose refinement, the corresponding weights for global translation, body pose, and 3D keypoints are 100, 10, and 10. In both stages, the temporal window weights are [0.5, 0.3, 0.1, 0.1].

A.3 Evaluation Metric Details

This appendix provides additional mathematical definitions and implementation details for the evaluation metrics introduced in Section 4.1. Detailed definitions are given where alignment or temporal evaluation choices affect the interpretation of the reported results. Standard image-quality metrics are documented together with the implementation settings used in the evaluation.

A.3.1 3D Reconstruction Metrics

Let $\hat{\mathbf{x}}_{t,p} \in \mathbb{R}^3$ and $\mathbf{x}_{t,p}^{\text{gt}} \in \mathbb{R}^3$ denote corresponding reconstructed and ground-truth 3D points, respectively, where t indexes the T matched frames and p indexes the P evaluated points. The common position error used for the vertex- and joint-based metrics is

$$E_{\text{pos}} = \frac{1000}{TP} \sum_{t=0}^{T-1} \sum_{p=1}^P \left\| \hat{\mathbf{x}}_{t,p} - \mathbf{x}_{t,p}^{\text{gt}} \right\|_2, \quad (12)$$

where the factor 1000 converts the underlying coordinates from meters to millimeters. The individual reconstruction metrics differ in the evaluated point representation and the alignment applied before computing E_{pos} .

MPVPE. For MPVPE, a single translation offset is estimated jointly over all T frames and K mesh vertices,

$$\boldsymbol{\delta} = \frac{1}{TK} \sum_{t=0}^{T-1} \sum_{k=1}^K \left(\mathbf{v}_{t,k}^{\text{gt}} - \hat{\mathbf{v}}_{t,k} \right). \quad (13)$$

The aligned reconstructed vertices are given by

$$\tilde{\mathbf{v}}_{t,k} = \hat{\mathbf{v}}_{t,k} + \boldsymbol{\delta}. \quad (14)$$

MPVPE corresponds to E_{pos} evaluated between $\tilde{\mathbf{v}}_{t,k}$ and $\mathbf{v}_{t,k}^{\text{gt}}$. Since a constant global translation offset does not affect the relative motion of the animation, it is considered irrelevant for the target use case and removed prior to evaluation. The same translation is applied to all frames, such that frame-dependent global translation, rotation, and scale errors are preserved.

Pelvis-Aligned MPJPE. Joint positions are obtained from the reconstructed and ground-truth mesh vertices using the gender-specific SMPL joint regressor. The implementation uses $J = 24$ SMPL joints. For each frame, the reconstructed joints are translated such that the reconstructed pelvis coincides with the ground-truth pelvis,

$$\boldsymbol{\delta}_t^{\text{pelvis}} = \mathbf{j}_{t,0}^{\text{gt}} - \hat{\mathbf{j}}_{t,0}, \quad (15)$$

where joint $j = 0$ denotes the pelvis. The aligned joints are therefore

$$\tilde{\mathbf{j}}_{t,j} = \hat{\mathbf{j}}_{t,j} + \boldsymbol{\delta}_t^{\text{pelvis}}. \quad (16)$$

Pelvis-aligned MPJPE corresponds to E_{pos} evaluated on these aligned joints. This removes frame-wise global translation while retaining differences in global orientation, scale, and articulated pose.

Procrustes-Aligned Metrics. PA-MPVPE and PA-MPJPE apply an independent similarity alignment to each frame. For reconstructed points $\widehat{\mathbf{X}}_t$ and corresponding ground-truth points $\mathbf{X}_t^{\text{gt}}$, the alignment determines a uniform scale s_t , a rotation $\mathbf{R}_t$, and a translation $\mathbf{d}_t$ that minimize

$$\left\| s_t \widehat{\mathbf{X}}_t \mathbf{R}_t^\top + \mathbf{1} \mathbf{d}_t^\top - \mathbf{X}_t^{\text{gt}} \right\|_F^2. \quad (17)$$

The transformation is estimated using singular value decomposition, with the rotation constrained to prevent reflections. PA-MPVPE applies this alignment to the mesh vertices, whereas PA-MPJPE applies it to the regressed SMPL joints. The common position error E_{pos} is then evaluated on the aligned points. Both metrics therefore remove translation, rotation, and uniform scale independently for each frame.

Acceleration Error. Temporal reconstruction error is computed from the pelvis-aligned reconstructed joints. The discrete second-order temporal difference for joint j is

$$\mathbf{a}_{t,j} = \mathbf{j}_{t+1,j} - 2\mathbf{j}_{t,j} + \mathbf{j}_{t-1,j}. \quad (18)$$

Acceleration error is the mean Euclidean difference between the second-order differences of the pelvis-aligned reconstructed joints and the corresponding ground-truth joints, averaged over center frames $t = 2, \dots, T - 3$. The temporal differences are not rescaled by the squared frame rate; acceleration error is therefore reported in mm/frame².

A.3.2 Synthesized Target-View Quality Metrics

For target-view quality evaluation, each synthesized frame $\widehat{I}_{t,i}$ is compared with the corresponding source-rendered frame $I_{t,i}^{\text{src}}$. Video frames are converted from BGR to RGB and normalized from $[0, 255]$ to $[0, 1]$ before evaluation. Each metric is computed for corresponding frame pairs and subsequently averaged over the frames of each target-view video.

PSNR. Peak Signal-to-Noise Ratio (PSNR) measures pixel-level fidelity based on the mean squared difference between corresponding images. It is computed using the TorchMetrics implementation with `data_range=1.0`, corresponding to the normalized image range. PSNR is reported in decibels (dB), with higher values indicating greater pixel-level agreement.

SSIM. Structural Similarity Index Measure (SSIM) evaluates image similarity in terms of local luminance, contrast, and structural information. It is computed using the TorchMetrics implementation with `data_range=1.0`; the remaining parameters use the library defaults. Higher SSIM values indicate greater structural similarity.

LPIPS. Learned Perceptual Image Patch Similarity (LPIPS) evaluates perceptual similarity by comparing feature representations extracted by a pretrained neural network rather than directly comparing RGB values. The evaluation uses TorchMetrics’ `LearnedPerceptualImagePatchSimilarity` with an AlexNet backbone (`net_type="alex"`) and input normalization enabled (`normalize=True`). Lower LPIPS values indicate greater perceptual similarity.

Declaration of AI Use

OpenAI ChatGPT (GPT-5.5/GPT-5.6 Sol) was used as an assistive tool during the preparation of this report. Its use was primarily limited to linguistic support, particularly for translating and refining text originally written in German. ChatGPT was also used to a limited extent for programming assistance, mainly for auxiliary utility scripts. All AI-assisted content and code suggestions were critically reviewed and verified before use.

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